

Assignment 2

3201NSC ADVANCED DIFFERENTIAL EQUATIONS AND MODELLING

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Question 1

(a) Solving the second-order differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = 0$$

with initial conditions $y(0) = 0$ and $y'(0) = 0$

Assume solution $y = e^{kx}$

$$y = e^{kx}$$

$$\frac{dy}{dx} = ke^{kx}$$

$$\frac{d^2y}{dx^2} = k^2e^{kx}$$

$$k^2e^{kx} + ke^{kx} - 2e^{kx} = 0$$

$$k^2 + k - 2 = 0$$

$$k = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$k = \frac{-(1) \pm \sqrt{(1)^2 - 4(1)(-2)}}{2(1)}$$

$$k = \frac{-1 \pm \sqrt{1+8}}{2}$$

$$k = \frac{-1 \pm \sqrt{9}}{2}$$

$$k = \frac{-1 \pm 3}{2}$$

$$k = \frac{-1+3}{2} \quad \text{or} \quad k = \frac{-1-3}{2}$$

$$k = 1 \quad \text{or} \quad k = -2$$

General Solution

$$y = Ae^x + Be^{-2x}$$

Applying initial conditions $y(0) = 0$ and $y'(0) = 0$

$$y(x) = Ae^x + Be^{-2x}$$

$$0 = Ae^{(0)} + Be^{-2(0)}$$

$$A + B = 0$$

$$B = -A$$

$$y'(x) = Ae^x - 2Be^{-2x}$$

$$0 = Ae^{(0)} - 2Be^{-2(0)}$$

$$A - 2B = 0$$

$$A - 2(-A) = 0$$

$$A + 2A = 0$$

$$3A = 0$$

$$A = 0$$

$$\Rightarrow B = 0$$

$$\therefore y = 0$$

(b) Deriving the Green's function that could be used to solve a differential equation of the form

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = f(x)$$

with initial conditions $y(0) = 0$ and $y'(0) = 0$

Step 1: Set up the differential equation for the Green's function

$$L_x G(x, t) = \delta(x - t)$$

The differential equation for $G(x, t)$ is therefore,

$$\frac{d^2 G(x, t)}{dx^2} + \frac{dG(x, t)}{dx} - 2G(x, t) = \delta(x - t)$$

with initial conditions $G(0, t) = 0$ and $G'(0, t) = 0$

Step 2: Look at the differential equation when x is below t

$$\frac{d^2 G(x, t)}{dx^2} + \frac{dG(x, t)}{dx} - 2G(x, t) = \delta(x - t)$$

The Dirac Delta function is zero where x is different from t

$$\therefore \frac{d^2 G(x, t)}{dx^2} + \frac{dG(x, t)}{dx} - 2G(x, t) = 0$$

The solution here is therefore

$$G(x, t) = Ae^x + Be^{-2x} \quad (\text{from part (a)})$$

(where A and B are some constants) when $x < t$

Step 3: Look at the differential equation when x is above t

$$\frac{d^2 G(x, t)}{dx^2} + \frac{dG(x, t)}{dx} - 2G(x, t) = \delta(x - t)$$

The Dirac Delta function is zero where x is different from t

$$\therefore \frac{d^2 G(x, t)}{dx^2} + \frac{dG(x, t)}{dx} - 2G(x, t) = 0$$

The solution is therefore

$$G(x, t) = Ce^x + De^{-2x} \quad (\text{from part (a)})$$

(where C and D are some other constants) when $x > t$

Step 4: Use the initial conditions

The solution for $G(x, t)$ when x is below t is

$$G(x, t) = Ae^x + Be^{-2x}$$

The initial conditions are that $G(0, t) = 0$ and $G'(0, t) = 0$

$\therefore$ For $x < t$, $G(x, t) = 0$ (from part (a))

Step 5: Join the two $G(x, t)$ solutions together at $x = t$

At $x = t$, $G(x, t)$ is continuous as x passes through t

The continuity of $G(x, t)$ as x passes from below t to above t means that

$$G(x, t) = G(x, t)$$

$$Ce^t + De^{-2t} = 0 \dots \textcircled{1}$$

Integrating the differential equation from $x = t - 0$ to $x = t + 0$

$$\int_{t-0}^{t+0} \left(\frac{d^2 G}{dx^2} + \frac{dG}{dx} - 2G \right) dx = \int_{t-0}^{t+0} \delta(x - t) dx$$

As we are integrating a continuous function across an interval which effectively has zero width, $\int_{t-0}^{t+0} G(x, t) dx = 0$

$$\left. \frac{dG}{dx} \right|_{t+0,t} - \left. \frac{dG}{dx} \right|_{t-0,t} + G|_{t+0,t} - G|_{t-0,t} - 2(0) = 1$$

$$\therefore Ce^t - 2De^{-2t} - 0 + Ce^t + De^{-2t} - 0 - 0 = 1$$

$$e^t(C + C) + e^{-2t}(-2D + D) = 1$$

$$2Ce^t - De^{-2t} = 1 \dots \textcircled{2}$$

Solving $\textcircled{1}$ and $\textcircled{2}$ simultaneously

$$Ce^t + De^{-2t} = 0$$

$$2Ce^t - De^{-2t} = 1$$

$\textcircled{1} + \textcircled{2}$ to eliminate D

$$3Ce^t = 1$$

$$C = \frac{1}{3}e^{-t}$$

Substitute $C = \frac{1}{3}e^{-t}$ into ① to solve for D

$$\left(\frac{1}{3}e^{-t}\right)e^t + De^{-2t} = 0$$

$$\frac{1}{3} + De^{-2t} = 0$$

$$De^{-2t} = -\frac{1}{3}$$

$$D = -\frac{1}{3}e^{2t}$$

$$G(x, t) = Ce^x + De^{-2x}$$

$$G(x, t) = \left(\frac{1}{3}e^{-t}\right)e^x + \left(-\frac{1}{3}e^{2t}\right)e^{-2x}$$

$$G(x, t) = \frac{1}{3}e^{-t+x} - \frac{1}{3}e^{2t-2x} \quad \text{for } x > t$$

Step 6: Conclusion

The Green's function for this differential equation is

$$\therefore G(x, t) = \begin{cases} 0 & x < t \\ \frac{1}{3}e^{(x-t)} - \frac{1}{3}e^{2(t-x)} & x > t \end{cases}$$

Step 7: Final Solution

The solution of the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = f(x)$$

with $y(0) = 0$ and $y'(0) = 0$ is

$$\therefore y(x) = \int_0^\infty G(x, t) f(t) dt$$

This solution can be split into

$$y(x) = \int_0^x G(x, t) f(t) dt + \int_x^\infty G(x, t) f(t) dt$$

$$y(x) = \int_0^x G(x, t) f(t) dt + \int_x^\infty 0 f(t) dt$$

$$y(x) = \int_0^x G(x, t) f(t) dt$$

$$y(x) = \int_0^x \left(\frac{1}{3} e^{x-t} - \frac{1}{3} e^{2(t-x)} \right) f(t) dt$$

$$y(x) = \frac{1}{3} \int_0^x (e^{(x-t)} - e^{2(t-x)}) f(t) dt$$

This form of the solution can be used to solve the differential equation with an arbitrary RHS function, $f(x)$.

(c) Using the Green's function found in part (b) to solve the differential equation

$$\frac{d^2 y}{dx^2} + \frac{dy}{dx} - 2y = e^{-x}$$

with initial conditions $y(0) = 0$ and $y'(0) = 0$

Here the RHS function is $f(x) = e^{-x}$, so using the Green's function

$$y(x) = \int_0^x G(x, t) e^{-t} dt$$

$$\begin{aligned} y(x) &= \int_0^x \left[\frac{1}{3} e^{(x-t)} - \frac{1}{3} e^{2(t-x)} \right] e^{-t} dt \\ &= \int_0^x \left(\frac{1}{3} e^{(x-2t)} - \frac{1}{3} e^{t-2x} \right) dt \\ &= \frac{1}{3} e^x \int_0^x e^{-2t} dt - \frac{1}{3} e^{-2x} \int_0^x e^t dt \\ &= \frac{1}{3} e^x \left[-\frac{1}{2} e^{-2t} \right]_0^x - \frac{1}{3} e^{-2x} [e^t]_0^x \\ &= \frac{1}{3} e^x \left\{ \left[-\frac{1}{2} e^{-2(x)} \right] - \left[-\frac{1}{2} e^{-2(0)} \right] \right\} - \frac{1}{3} e^{-2x} \{ [e^x] - [e^0] \} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{3}e^x \left[-\frac{1}{2}e^{-2x} + \frac{1}{2} \right] - \frac{1}{3}e^{-2x}[e^x - 1] \\
&= -\frac{1}{6}e^{-x} + \frac{1}{6}e^x - \frac{1}{3}e^{-x} + \frac{1}{3}e^{-2x} \\
\therefore y(x) &= \frac{1}{6}e^x + \frac{1}{3}e^{-2x} - \frac{1}{2}e^{-x}
\end{aligned}$$

(d) Using the answer from part (c) to solve the differential equation

$$\frac{d^2y}{dx^2} + \frac{dy}{dx} - 2y = e^{-x}$$

with initial conditions $y(0) = 0$ and $y'(0) = -1$

To solve this differential equation with nonzero initial conditons, let

$y(x) = u(x) + v(x)$ where

$$\frac{d^2u}{dx^2} + \frac{du}{dx} - 2u = e^{-x} \quad \text{with } u(0) = 0 \quad \text{and } u'(0) = 0$$

$$\frac{d^2v}{dx^2} + \frac{dv}{dx} - 2v = 0 \quad \text{with } v(0) = 1 \quad \text{and } v'(0) = -1$$

From part (c), the solution to u is:

$$u(x) = \frac{1}{6}e^x + \frac{1}{3}e^{-2x} - \frac{1}{2}e^{-x}$$

and from part (a), the solution for v is:

$$v(x) = Ee^x + Fe^{-2x}$$

$$v(0) = 1 \implies 1 = Ee^{(0)} + Fe^{-2(0)}$$

$$1 = E + F$$

$$F = 1 - E$$

$$v'(0) = -1 \implies -1 = Ee^{(0)} - 2Fe^{-2(0)}$$

$$-1 = E - 2F$$

$$-1 = E - 2(1 - E)$$

$$-1 = E - 2 + 2E$$

$$3E = 1$$

$$E = \frac{1}{3}$$

$$F = 1 - \frac{1}{3}$$

$$F = \frac{2}{3}$$

$$\therefore v(x) = \frac{1}{3}e^x + \frac{2}{3}e^{-2x}$$

$$\therefore y(x) = u(x) + v(x)$$

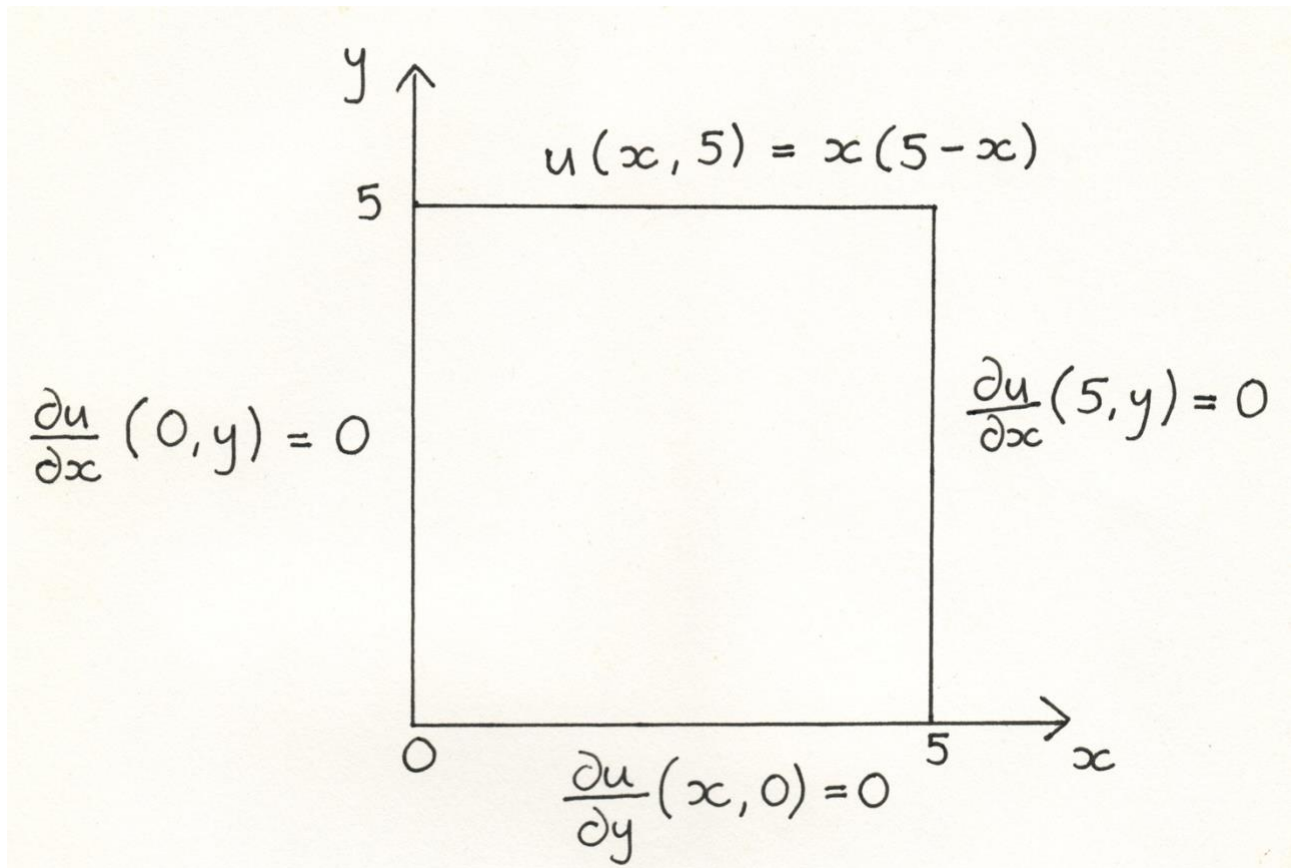
$$= \frac{1}{6}e^x + \frac{1}{3}e^{-2x} - \frac{1}{2}e^{-x} + \frac{1}{3}e^x + \frac{2}{3}e^{-2x}$$

$$= \left(\frac{1}{6} + \frac{1}{3}\right)e^x + \left(\frac{1}{3} + \frac{2}{3}\right)e^{-2x} - \frac{1}{2}e^{-x}$$

$$\therefore y(x) = \frac{1}{2}e^x + e^{-2x} - \frac{1}{2}e^{-x}$$

Question 2

- (a) The determined appropriate boundary conditions for $u(x, y)$ are included on the diagram below. The coordinate axes are as shown in the diagram, with the origin in the lower left-hand corner, the x-axis horizontal and the y-axis vertical (since the plate is square).



- (b) Assuming that the solution for the temperature within the object, $u(x, y)$ can be written in the form

$$u(x, y) = X(x)Y(y)$$

Determining appropriate boundary conditions for $X(x)$ and $Y(y)$

Rewriting the Boundary Conditions

$$\frac{\partial u}{\partial y}(x, 0) = 0, \quad X(x) \frac{dY}{dy}(0) = 0 \Rightarrow \frac{dY}{dy}(0) = 0$$

Along the bottom edge, $\frac{dY}{dy}(0) = 0$ as $X(x)$ would be zero for all values of x , leading to a trivial solution for $u(x, y) = X(x)Y(y)$ across the entire plate

$$\frac{\partial u}{\partial x}(0, y) = 0, \quad \frac{dX}{dx}(0)Y(y) = 0 \Rightarrow \frac{dX}{dx}(0) = 0$$

Along the left edge, $\frac{dX}{dx}(0) = 0$ as $Y(y)$ would be zero for all values of y , leading to a trivial solution for $u(x, y) = X(x)Y(y)$ across the entire plate

$$\frac{\partial u}{\partial x}(5, y) = 0, \quad \frac{dX}{dx}(5)Y(y) = 0 \Rightarrow \frac{dX}{dx}(5) = 0$$

Along the right edge, $\frac{dX}{dx}(5) = 0$ as $Y(y)$ would be zero for all values of y , leading to a trivial solution for $u(x, y) = X(x)Y(y)$ across the entire plate

$$u(x, 5) = x(5 - x), \quad X(x)Y(5) = x(5 - x)$$

Along the top edge, there is no useful information for $X(x)$ and $Y(y)$

- (c) It is assumed that the temperature of the plate can be accurately modelled using Laplace's equation. Moreover, since the plate is square and thin:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

where $u(x, y)$ is the temperature at any point (x, y)

Substituting $u(x, y) = X(x)Y(y)$ into Laplace's Equation and deriving differential equations for $X(x)$ and $Y(y)$

$$\frac{\partial^2 X(x)Y(y)}{\partial x^2} + \frac{\partial^2 X(x)Y(y)}{\partial y^2} = 0$$

$$Y(y)\frac{\partial^2 X(x)}{\partial x^2} + X(x)\frac{\partial^2 Y(y)}{\partial y^2} = 0$$

$$\frac{d^2 X}{dx^2}Y + X\frac{d^2 Y}{dy^2} = 0$$

$$\frac{\frac{d^2 X}{dx^2}Y}{XY} + \frac{X\frac{d^2 Y}{dy^2}}{XY} = 0$$

$$\frac{1}{X}\frac{d^2 X}{dx^2} + \frac{1}{Y}\frac{d^2 Y}{dy^2} = 0$$

$$\frac{1}{X} \frac{d^2 X}{dx^2} = -\frac{1}{Y} \frac{d^2 Y}{dy^2}$$

Since the LHS is a function of x only and the RHS is a function of y only, both must be a constant

$$\therefore \frac{1}{X} \frac{d^2 X}{dx^2} = -\frac{1}{Y} \frac{d^2 Y}{dy^2} = C$$

where C is a constant

$$\frac{1}{X} \frac{d^2 X}{dx^2} = C$$

$$-\frac{1}{Y} \frac{d^2 Y}{dy^2} = C$$

$$\frac{d^2 X}{dx^2} = CX$$

$$\frac{d^2 Y}{dy^2} = -CY$$

Thus, the separated differential equations for $X(x)$ and $Y(y)$ are

$$\frac{d^2 X}{dx^2} - CX = 0 \quad \text{and} \quad \frac{d^2 Y}{dy^2} + CY = 0$$

- (d) Solving for $X(x)$ and $Y(y)$, ensuring that all three cases for the separation constant (positive, negative, zero) are considered.

Assume C is **positive**. Let $C = k^2$

$$\frac{d^2 X}{dx^2} - k^2 X = 0 \quad \text{and} \quad \frac{d^2 Y}{dy^2} + k^2 Y = 0$$

The solutions for these equations are

$$\begin{aligned} X(x) &= A \cosh(kx) + B \sinh(kx) & \text{and} & & Y(y) &= D \cos(ky) + E \sin(ky) \\ \therefore \frac{dX}{dx} &= kA \sinh(kx) + kB \cosh(kx) & & & \frac{dY}{dy} &= -kD \sin(ky) + kE \cos(ky) \end{aligned}$$

$$\frac{dX}{dx}(0) = 0 \Rightarrow 0 = kA \sinh(k \cdot 0) + kB \cosh(k \cdot 0)$$

$$\Rightarrow B = 0$$

$$\frac{dX}{dx}(5) = 0 \Rightarrow 0 = kA \sinh(5k)$$

$$\Rightarrow A = 0$$

$\therefore$ Trivial solution. $X(x) = 0$ for all values of x , which leads to $u(x, y) = 0$
 $\Rightarrow C$ is not positive

Assume C is **zero**. Let $C = 0$

$$\frac{d^2X}{dx^2} = 0 \quad \text{and} \quad \frac{d^2Y}{dy^2} = 0$$

The solutions for these equations are

$$X(x) = Ax + B \quad \text{and} \quad Y(y) = Dy + E$$

$$\therefore \frac{dX}{dx} = A \quad \frac{dY}{dy} = D$$

$$\frac{dX}{dx}(0) = 0 \Rightarrow A = 0 \quad \frac{dY}{dy}(0) = 0 \Rightarrow D = 0$$

$$\therefore X(x) = B \quad \text{and} \quad Y(y) = E$$

$\therefore C = 0$ is a possible solution

Assume C is **negative**. Let $C = -k^2$

$$\frac{d^2X}{dx^2} + k^2X = 0 \quad \text{and} \quad \frac{d^2Y}{dy^2} - k^2Y = 0$$

The solutions for these equations are

$$X(x) = A \cos(kx) + B \sin(kx) \quad \text{and} \quad Y(y) = D \sinh(ky) + E \cosh(ky)$$

$$\therefore \frac{dX}{dx} = -kA \sin(kx) + kB \cos(kx) \quad \frac{dY}{dy} = kD \cosh(ky) + kE \sinh(ky)$$

$$\frac{dX}{dx}(0) = 0 \Rightarrow 0 = -kA \sin(k \cdot 0) + kB \cos(k \cdot 0)$$

$$\Rightarrow B = 0$$

$$\frac{dX}{dx}(5) = 0 \Rightarrow -kA \sin(5k) = 0$$

$$\Rightarrow 5k = n\pi \quad n = 1, 2, 3 \dots$$

$$\Rightarrow k = \frac{n\pi}{5}$$

$$\frac{dY}{dy}(0) = 0 \Rightarrow 0 = kD \cosh(k \cdot 0) + kE \sinh(k \cdot 0)$$

$$0 = kD + 0$$

$$\Rightarrow D = 0$$

$$\therefore X(x) = A \cos\left(\frac{n\pi x}{5}\right) \quad \text{and} \quad Y(y) = E \cosh\left(\frac{n\pi y}{5}\right)$$

$\therefore C$ can be zero and negative

(e) Therefore, the solution is given by

$$u(x, y) = X(x)Y(y)$$

$$\therefore u_n(x, y) = F_n \cos\left(\frac{n\pi x}{5}\right) \cosh\left(\frac{n\pi y}{5}\right) \quad n = 1, 2, 3 \dots$$

and the general solution for $u(x, y)$ as an infinite series can be written as

$$u(x, y) = u_0(x, y) + \sum_{n=1}^{\infty} u_n(x, y)$$

$$u(x, y) = F_0 + \sum_{n=1}^{\infty} F_n \cos\left(\frac{n\pi x}{5}\right) \cosh\left(\frac{n\pi y}{5}\right)$$

- (f) To determine the unknown coefficients F_0 and F_n , the non-zero remaining boundary condition $u(x, 5) = x(5 - x)$ is used.

$$x(5 - x) = F_0 + \sum_{n=1}^{\infty} F_n \cos\left(\frac{n\pi x}{5}\right) \cosh\left(\frac{n\pi 5}{5}\right)$$

$$x(5 - x) = F_0 + \sum_{n=1}^{\infty} F_n \cos\left(\frac{n\pi x}{5}\right) \cosh(n\pi)$$

Exploiting the orthogonality of the cosine function by multiplying both sides of the equation by $\cos\left(\frac{m\pi x}{5}\right)$ and integrating over the side length of the plate, from 0 to 5.

$$\int_0^5 x(5 - x) \cos\left(\frac{m\pi x}{5}\right) dx = \int_0^5 F_0 \cos\left(\frac{m\pi x}{5}\right) dx + \sum_{n=1}^{\infty} F_n \cos\left(\frac{n\pi x}{5}\right) \cosh(n\pi) \cos\left(\frac{m\pi x}{5}\right) dx$$

$$\int_0^5 x(5 - x) \cos\left(\frac{m\pi x}{5}\right) dx = \int_0^5 F_0 \cos\left(\frac{m\pi x}{5}\right) dx + \sum_{n=1}^{\infty} F_n \cosh(n\pi) \int_0^5 \cos\left(\frac{n\pi x}{5}\right) \cos\left(\frac{m\pi x}{5}\right) dx$$

By parts for $m > 0$

Using integration by parts tabular method

signs	$u \rightarrow \frac{d}{dx}$	$dv \rightarrow \int dx$
+	$x(5 - x) \Rightarrow 5x - x^2$	$\cos\left(\frac{m\pi x}{5}\right)$
-	$5 - 2x$	$\frac{5}{m\pi} \sin\left(\frac{m\pi x}{5}\right)$
+	-2	$-\frac{25}{m^2\pi^2} \cos\left(\frac{m\pi x}{5}\right)$
-	0	$-\frac{125}{m^3\pi^3} \sin\left(\frac{m\pi x}{5}\right)$

$$LHS = \left[x(5 - x) \cdot \frac{5}{m\pi} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 - \left[(5 - 2x) \cdot -\frac{25}{m^2\pi^2} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 + \left[-2 \cdot -\frac{125}{m^3\pi^3} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5$$

$$LHS = \left[\frac{5x(5 - x)}{m\pi} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 + \left[\frac{25(5 - 2x)}{m^2\pi^2} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 + \left[\frac{250}{m^3\pi^3} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5$$

Evaluating each integral seperately

$$\begin{aligned}\left[\frac{5x(5-x)}{m\pi}\sin\left(\frac{m\pi x}{5}\right)\right]_0^5 &= \left\{\left[\frac{5 \cdot 5(5-5)}{m\pi}\sin\left(\frac{m\pi \cdot 5}{5}\right)\right] - \left[-\frac{5 \cdot 0(5-0)}{m\pi}\sin\left(\frac{m\pi \cdot 0}{5}\right)\right]\right\} \\ &= \{[0] - [0]\} \\ &= 0\end{aligned}$$

$$\begin{aligned}\left[\frac{25(5-2x)}{m^2\pi^2}\cos\left(\frac{m\pi x}{5}\right)\right]_0^5 &= \left\{\left[\frac{25(5-2 \cdot 5)}{m^2\pi^2}\cos\left(\frac{m\pi \cdot 5}{5}\right)\right] - \left[\frac{25(5-2 \cdot 0)}{m^2\pi^2}\cos\left(\frac{m\pi \cdot 0}{5}\right)\right]\right\} \\ &= \left\{\left[\frac{25(-5)}{m^2\pi^2}\cos(m\pi)\right] - \left[\frac{25(5)}{m^2\pi^2}\cos(0)\right]\right\} \\ &= \left\{\left[-\frac{125}{m^2\pi^2}\cos(m\pi)\right] - \left[\frac{125}{m^2\pi^2} \times 1\right]\right\} \\ &= -\frac{125}{m^2\pi^2}\cos(m\pi) - \frac{125}{m^2\pi^2}\end{aligned}$$

$$\begin{aligned}\left[\frac{250}{m^3\pi^3}\sin\left(\frac{m\pi x}{5}\right)\right]_0^5 &= \left\{\left[\frac{250}{m^3\pi^3}\sin\left(\frac{m\pi \cdot 5}{5}\right)\right] - \left[\frac{250}{m^3\pi^3}\sin\left(\frac{m\pi \cdot 0}{5}\right)\right]\right\} \\ &= \left\{\left[\frac{250}{m^3\pi^3}\sin(m\pi)\right] - [0]\right\} \\ &= \frac{250}{m^3\pi^3}\sin(m\pi)\end{aligned}$$

$$\begin{aligned}\therefore LHS &= 0 - \frac{125}{m^2\pi^2}\cos(m\pi) - \frac{125}{m^2\pi^2} + \frac{250}{m^3\pi^3}\sin(m\pi) \\ &= -\frac{125}{m^2\pi^2} - \frac{125}{m^2\pi^2}\cos(m\pi) + \frac{250}{m^3\pi^3}\sin(m\pi) \\ LHS &= -\frac{125}{m^2\pi^2}\left[1 + \cos(m\pi) - \frac{2}{m\pi}\sin(m\pi)\right]\end{aligned}$$

For m even

$$\begin{aligned} LHS &= -\frac{125}{m^2\pi^2} [1 + 1 - 0] \\ &= -\frac{125}{m^2\pi^2} (2) \\ &= -\frac{250}{m^2\pi^2} \end{aligned}$$

For m odd

$$\begin{aligned} LHS &= -\frac{125}{m^2\pi^2} [1 - 1 - 0] \\ &= -\frac{125}{m^2\pi^2} (0) \\ &= 0 \end{aligned}$$

$$\therefore LHS = \begin{cases} -\frac{250}{m^2\pi^2} & m \text{ even} \\ 0 & m \text{ odd} \end{cases} \quad m > 0$$

For $m > 0$ and m even,

$$RHS = F_m \cosh(m\pi) \times \frac{5}{2} = -\frac{250}{m^2\pi^2} = LHS$$

$$\therefore \text{For } m > 0, F_m = \begin{cases} -\frac{100}{m^2\pi^2 \cosh(m\pi)} & m \text{ even} \\ 0 & m \text{ odd} \end{cases}$$

and for $m = 0$,

$$\begin{aligned} \int_0^5 5x - x^2 dx &= \int_0^5 F_0 dx + \sum_{n=1}^{\infty} F_n \cosh(n\pi) \int_0^5 \cos\left(\frac{n\pi x}{5}\right) dx \\ &= \left[\frac{5x^2}{2} - \frac{x^3}{3} \right]_0^5 = [F_0 x]_0^5 + 0 \end{aligned}$$

$$\left[\frac{5(5)^2}{2} - \frac{(5)^3}{3} \right] - \left[\frac{5(0)^2}{2} - \frac{(0)^3}{3} \right] = [F_0(5)] - [F_0(0)]$$

$$\frac{125}{6} = 5F_0$$

$$F_0 = \frac{25}{6}$$

Therefore, the final solution for $u(x, y)$ can be written as

$$u(x, y) = F_0 + \sum_{n=1}^{\infty} F_n \cos\left(\frac{n\pi x}{5}\right) \cosh\left(\frac{n\pi y}{5}\right)$$

$$\therefore u(x, y) = \frac{25}{6} - \sum_{\substack{n=1 \\ n \text{ even}}}^{\infty} \frac{100}{m^2 \pi^2 \cosh(m\pi)} \cos\left(\frac{m\pi x}{5}\right) \cosh\left(\frac{m\pi y}{5}\right)$$

Question 3

- (a) Based on the physical properties, assuming the drumhead is very thin compared to its radius, the drumhead has no z dimension and the term $\frac{\partial^2 u}{\partial z^2}$ can thus be ignored.

Therefore, the wave equation in cylindrical coordinates

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} + \frac{\partial^2 u}{\partial z^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

can be reduced to

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$$

- (b) The boundary and initial conditions for u are determined below.

At $r = 20$ (cylinder surface = boundary) the drumhead is fixed (around its circumference). Therefore, the displacement equals 0 there.

$$\therefore u(20, \theta, t) = 0$$

At $t = 0$, the displacement is 0.

$$\therefore u(r, \theta, 0) = 0$$

At $t = 0$, the initial velocity (i.e., $\frac{\partial u}{\partial t}$) is $0.01(20 - r) \sin(3\theta)$

$$\therefore \frac{\partial u}{\partial t}(r, \theta, 0) = 0.01(20 - r) \sin(3\theta)$$

- (c) It is assumed that the wave equation can be solved using separation of variables where u can be written in the form

$$u(r, \theta, t) = R(r)\Theta(\theta)T(t)$$

The function $\Theta(\theta)$ must be a periodic function of the angle θ ; in other words, the solution is required to be periodic with respect to θ . For example, if θ is increased by 2π , it should not change the solution.

$$u(r, \theta, t) = u(r, \theta + 2\pi, t)$$

More specifically, the angle θ wraps around the drumhead from 0 to 2π . When θ reaches 2π it has come back to the same place as $\theta = 0$. The value of u when $\theta = 2\pi$ is the same as the value of u when $\theta = 0$. The simplest functions that do this are basic periodic functions such as: $\sin \theta$, $\cos \theta$, $\sin 2\theta$, $\cos 2\theta$ etc., which all have 2π as one of their periods.

Therefore, it is assumed that $\Theta(\theta)$ is either of the form:

$$\sin(n\theta) \quad \text{or} \quad \cos(n\theta)$$

where n must be an integer (0, 1, 2 etc.)

Choosing $\sin(n\theta)$ due to the B.C $\frac{\partial u}{\partial t}(r, \theta, 0) = 0.01(20 - r) \sin(3\theta)$

Moreover, choosing $n = 3$ to match the $\sin(3\theta)$ term from the B.C

$$\therefore \Theta(\theta) = \sin(3\theta)$$

$$\therefore u(r, \theta, t) = R(r) \sin(3\theta) T(t)$$

Rewriting the Boundary Conditions

$$u(20, \theta, t) = 0, \quad R(20) \sin(3\theta) T(t) = 0 \Rightarrow R(20) = 0$$

$$u(r, \theta, 0) = 0, \quad R(r) \sin(3\theta) T(0) = 0 \Rightarrow T(0) = 0$$

$$\frac{\partial u}{\partial t}(r, \theta, 0) = 0.01(20 - r) \sin(3\theta), \quad R(r) \sin(3\theta) \frac{dT}{dt}(0) = 0.01(20 - r) \sin(3\theta)$$

No useful information

- (d) Substituting $u(r, \theta, t) = R(r) \sin(3\theta) T(t)$ into the wave equation to derive differential equations for $R(r)$ and $T(t)$

$$\begin{aligned} \frac{\partial^2 R(r) \sin(3\theta) T(t)}{\partial r^2} + \frac{1}{r} \frac{\partial R(r) \sin(3\theta) T(t)}{\partial r} + \frac{1}{r^2} \frac{\partial^2 R(r) \sin(3\theta) T(t)}{\partial \theta^2} &= \frac{1}{c^2} \frac{\partial^2 R(r) \sin(3\theta) T(t)}{\partial t^2} \\ \sin(3\theta) T(t) \frac{\partial^2 R(r)}{\partial r^2} + \sin(3\theta) T(t) \frac{1}{r} \frac{\partial R(r)}{\partial r} + R(r) T(t) \frac{1}{r^2} \frac{\partial^2 \sin(3\theta)}{\partial \theta^2} &= R(r) \sin(3\theta) \frac{1}{c^2} \frac{\partial^2 T(t)}{\partial t^2} \\ R'' \sin(3\theta) T + \frac{1}{r} R' \sin(3\theta) T + \frac{1}{r^2} R [-9 \sin(3\theta)] T &= \frac{1}{c^2} R \sin(3\theta) T'' \\ R'' \sin(3\theta) T + \frac{1}{r} R' \sin(3\theta) T - \frac{9}{r^2} R \sin(3\theta) T &= \frac{1}{c^2} R \sin(3\theta) T'' \\ \frac{R'' \sin(3\theta) T}{R \sin(3\theta) T} + \frac{\frac{1}{r} R' \sin(3\theta) T}{R \sin(3\theta) T} - \frac{\frac{9}{r^2} R \sin(3\theta) T}{R \sin(3\theta) T} &= \frac{\frac{1}{c^2} R \sin(3\theta) T''}{R \sin(3\theta) T} \\ \frac{R''}{R} + \frac{1}{r} \frac{R'}{R} - \frac{9}{r^2} &= \frac{1}{c^2} \frac{T''}{T} \end{aligned}$$

Since the LHS is a function that only depends on the variable r and the RHS is a function that only depends on the variable t , the only way that this can happen is that both sides are the same constant.

$$\therefore \frac{R''}{R} + \frac{1}{r} \frac{R'}{R} - \frac{9}{r^2} = \frac{1}{c^2} \frac{T''}{T} = \lambda$$

where λ is a constant

$$\frac{R''}{R} + \frac{1}{r} \frac{R'}{R} - \frac{9}{r^2} = \lambda$$

$$\frac{1}{c^2} \frac{T''}{T} = \lambda$$

$$r^2 R'' + r R' - 9R = \lambda r^2 R$$

$$T'' = \lambda c^2 T$$

Thus, the separated differential equations for $R(r)$ and $T(t)$ are

$$r^2 R'' + r R' - 9R - \lambda r^2 R = 0$$

and

$$T'' - \lambda c^2 T = 0$$

- (e) By considering the cases where the separation constant could be positive, negative, or zero, general solutions for $R(r)$ and $T(t)$ were determined.

Assume λ is **positive**. Let $\lambda = k^2$

$$r^2 R'' + rR' - (k^2 r^2 + 3^2)R = 0 \quad \text{and} \quad T'' - k^2 c^2 T = 0$$

The solutions for these equations are

$$R(r) = AI_3(kr) + BK_3(kr) \quad \text{and} \quad T(t) = D \sinh(kct) + E \cosh(kct)$$

Require u to be finite at $r = 0$

$$\therefore B = 0$$

$$R(20) = 0 \Rightarrow 0 = AI_3(20k)$$

$$\Rightarrow A = 0$$

$\therefore$ Trivial solution. $R(r) = 0$ for all values of r , which leads to $u(r, \theta, t) = 0$

$\Rightarrow \lambda$ is not positive

Assume λ is **zero**. Let $\lambda = 0$

$$r^2 R'' + rR' - 9R = 0 \quad \text{and} \quad T'' = 0$$

The solutions for these equations are

$$R(r) = Ar^3 + \frac{B}{r^3} \quad \text{and} \quad T(t) = Dt + E$$

Require u to be finite at $r = 0$

$$\therefore B = 0$$

$$R(20) = 0 \Rightarrow 0 = A \times (20)^3$$

$$0 = 8000A$$

$$\Rightarrow A = 0$$

$\therefore$ Trivial solution. $R(r) = 0$ for all values of r , which leads to $u(r, \theta, t) = 0$

$\Rightarrow \lambda$ is not positive

Assume λ is **negative**. Let $\lambda = -k^2$

$$r^2 R'' + rR' + (k^2 r^2 - 3^2)R = 0 \quad \text{and} \quad T'' + k^2 c^2 T = 0$$

The solutions for these equations are

$$R(r) = AJ_3(kr) + BY_3(kr) \quad \text{and} \quad T(t) = D \sin(kct) + E \cos(kct)$$

Require u to be finite at $r = 0$

$$\therefore B = 0$$

$$R(20) = 0 \Rightarrow 0 = AJ_3(20k)$$

$20k$ must be equal to a zero of J_3

Let $20k = \alpha_p$ where α_p is the p th zero of J_3

$$\Rightarrow k = \frac{\alpha_p}{20} \quad p = 1, 2, 3 \dots$$

$$T(0) = 0 \Rightarrow 0 = D \sin(kc \cdot 0) + E \cos(kc \cdot 0) \\ \Rightarrow E = 0$$

$$\therefore R(r) = AJ_3\left(\frac{\alpha_p}{20}r\right) \quad \text{and} \quad T(t) = D \sin\left(\frac{\alpha_p c}{20}t\right)$$

$\therefore \lambda$ is negative

(f) Therefore, the solution is given by

$$u(r, \theta, t) = R(r) \sin(3\theta) T(t)$$

$$\therefore u_p(r, \theta, t) = A_p J_3\left(\frac{\alpha_p}{20}r\right) \sin(3\theta) D_p \sin\left(\frac{\alpha_p c}{20}t\right) \quad p = 1, 2, 3 \dots$$

and the general solution for $u(x, y)$ as an infinite series can be written as

$$u(r, \theta, t) = \sum_{p=1}^{\infty} F_p J_3\left(\frac{\alpha_p}{20}r\right) \sin(3\theta) \sin\left(\frac{\alpha_p c}{20}t\right)$$

(g) To determine the unknown coefficient the remaining boundary condition,

$$\frac{\partial u}{\partial t}(r, \theta, 0) = 0.01(20 - r) \sin(3\theta) \text{ is used.}$$

$$\begin{aligned} \frac{\partial u}{\partial t} &= \sum_{p=1}^{\infty} F_p J_3\left(\frac{\alpha_p}{20}r\right) \sin(3\theta) \left[\frac{\alpha_p c}{20}\right] \cos\left(\frac{\alpha_p c}{20}t\right) \\ \therefore 0.01(20 - r) \sin(3\theta) &= \sum_{p=1}^{\infty} F_p J_3\left(\frac{\alpha_p}{20}r\right) \sin(3\theta) \left[\frac{\alpha_p c}{20}\right] \cos\left(\frac{\alpha_p c}{20} \cdot 0\right) \\ 0.01(20 - r) \sin(3\theta) &= \sum_{p=1}^{\infty} F_p J_3\left(\frac{\alpha_p}{20}r\right) \sin(3\theta) \left[\frac{\alpha_p c}{20}\right] \times 1 \\ 0.01(20 - r) &= \sum_{p=1}^{\infty} F_p \left(\frac{\alpha_p c}{20}\right) J_3\left(\frac{\alpha_p}{20}r\right) \end{aligned}$$

Exploiting the orthogonality relationship of the Bessel function, J_3 , by multiplying both sides of the equation by $rJ_3\left(\frac{\alpha_q}{20}r\right)$ and integrating over the radius of the drumhead, from 0 to 20.

$$\begin{aligned} \int_0^{20} 0.01(20 - r) r J_3\left(\frac{\alpha_q}{20}r\right) dr &= \int_0^{20} \sum_{p=1}^{\infty} F_p \left(\frac{\alpha_p c}{20}\right) J_3\left(\frac{\alpha_p}{20}r\right) r J_3\left(\frac{\alpha_q}{20}r\right) dr \\ \int_0^{20} 0.01r(20 - r) J_3\left(\frac{\alpha_q}{20}r\right) dr &= \sum_{p=1}^{\infty} F_p \left(\frac{\alpha_p c}{20}\right) \int_0^{20} r J_3\left(\frac{\alpha_p}{20}r\right) J_3\left(\frac{\alpha_q}{20}r\right) dr \\ \int_0^{20} 0.01r(20 - r) J_3\left(\frac{\alpha_q}{20}r\right) dr &= F_q \left(\frac{\alpha_q c}{20}\right) \times \frac{20^2}{2} J_4(\alpha_q) \\ \int_0^{20} 0.01r(20 - r) J_3\left(\frac{\alpha_q}{20}r\right) dr &= F_q \times \frac{\alpha_q c}{20} \times 200 J_4(\alpha_q) \\ \int_0^{20} 0.01r(20 - r) J_3\left(\frac{\alpha_q}{20}r\right) dr &= 10\alpha_q c J_4(\alpha_q) F_q \\ F_q &= \frac{1}{10\alpha_q c J_4(\alpha_q)} \int_0^{20} 0.01r(20 - r) J_3\left(\frac{\alpha_q}{20}r\right) dr \end{aligned}$$

$$\therefore u(r, \theta, t) = \sum_{q=1}^{\infty} F_q J_3\left(\frac{\alpha_q}{20}r\right) \sin(3\theta) \sin\left(\frac{\alpha_q c}{20}t\right)$$

where F_q is given above and

α_q are the zeroes of J_3

Appendix: Alternate Integration by Parts Calculation (Traditional Method) for Q2

$$\int u \, dv = uv - \int v \, du$$

$$\begin{array}{ll} \text{Let } u = x(5-x) & \downarrow \text{deriv} \\ du = (5-2x) \, dx & \end{array} \quad \begin{array}{ll} v = \frac{5}{m\pi} \sin\left(\frac{m\pi x}{5}\right) & \uparrow \text{integ} \\ dv = \cos\left(\frac{m\pi x}{5}\right) \, dx & \end{array}$$

$$\begin{aligned} \therefore \int_0^5 x(5-x) \cos\left(\frac{m\pi x}{5}\right) \, dx &= \left[x(5-x) \times \frac{5}{m\pi} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 - \int_0^5 \frac{5}{m\pi} \sin\left(\frac{m\pi x}{5}\right) \times (5-2x) \, dx \\ &= \left[\frac{5x(5-x)}{m\pi} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 - \underbrace{\frac{5}{m\pi} \int_0^5 (5-2x) \sin\left(\frac{m\pi x}{5}\right) \, dx}_{\text{integration by parts again}} \quad \textcircled{1} \end{aligned}$$

$$\begin{array}{ll} \text{Let } u = (5-2x) & \downarrow \text{deriv} \\ du = -2x \, dx & \end{array} \quad \begin{array}{ll} v = -\frac{5}{m\pi} \cos\left(\frac{m\pi x}{5}\right) & \uparrow \text{integ} \\ dv = \sin\left(\frac{m\pi x}{5}\right) \, dx & \end{array}$$

$$\begin{aligned} \therefore \int_0^5 (5-2x) \sin\left(\frac{m\pi x}{5}\right) \, dx &= \left[(5-2x) \times -\frac{5}{m\pi} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 - \int_0^5 -\frac{5}{m\pi} \cos\left(\frac{m\pi x}{5}\right) \times -2 \, dx \\ &= \left[-\frac{5(5-2x)}{m\pi} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 - \int_0^5 \frac{10}{m\pi} \cos\left(\frac{m\pi x}{5}\right) \, dx \\ &= \left[-\frac{5(5-2x)}{m\pi} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 - \frac{10}{m\pi} \int_0^5 \cos\left(\frac{m\pi x}{5}\right) \, dx \\ &= \left[-\frac{5(5-2x)}{m\pi} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 - \frac{10}{m\pi} \left[\frac{5}{m\pi} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 \\ &= \left[-\frac{5(5-2x)}{m\pi} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 - \left[\frac{50}{m^2\pi^2} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 \quad \textcircled{2} \end{aligned}$$

Substituting ② back into ①

$$\begin{aligned} \therefore \int_0^5 x(5-x) \cos\left(\frac{m\pi x}{5}\right) \, dx &= \left[\frac{5x(5-x)}{m\pi} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 - \frac{5}{m\pi} \left\{ \left[-\frac{5(5-2x)}{m\pi} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 - \left[\frac{50}{m^2\pi^2} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 \right\} \\ &= \left[\frac{5x(5-x)}{m\pi} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 + \left[\frac{25(5-2x)}{m^2\pi^2} \cos\left(\frac{m\pi x}{5}\right) \right]_0^5 + \left[\frac{250}{m^3\pi^3} \sin\left(\frac{m\pi x}{5}\right) \right]_0^5 \end{aligned}$$